

READYKIT · DESIGN DECISIONS REGISTER

ReadyKit design decisions register

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DOCUMENT CONTROL

Document ID	RDK-DEC-001	Version	0.4
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/readykit		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Value-engineering wording restated; follow-ups of the 2026-10-02 decisions carried into the model, BOM, calculations and pictures
0.3	2026-10-02	Amish Chadha	Design for construction (RDK-DDR-003, P1 to P6) accepted by Amish on 2026-10-02; the 2026-09-30 row no longer says open for review
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 6 (RDK-DDR-003 A1 and A2 decided); moved to decisions made; To confirm item 2 updated
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

ReadyKit design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	The pinned 3D viewer script's size (about 1 MB assumed) and its Apache-2.0 notice	It is copied into every repository's media folder beside the 3D model, and its notice must travel with it	RDK-DDR-003, P4
2	Pango and fontconfig install cleanly on macOS (Homebrew) and on Windows through WSL2, the route decided on 2026-10-02	PDF output needs them; R2 cannot be checked without them	RDK-REQ-001, R2
3	The license audit still holds for every library in every extra, including CairoSVG under LGPL-3.0 used as an installed dependency	R14 and the MIT license of the package	RDK-CAL-001, section F
4	The free CI runner images still provide the Pango packages the workflow installs	The release job renders PDFs on the runner	Build plan section 3.1
5	The core install really is about 5 MB in a clean environment (measured here from installed libraries, not from a fresh install)	Sets the setup time (R1) and the CI check time	RDK-DDR-003, P3

■ Value engineering

Value-engineering target: USD 0. Estimated cost of the constructable design: USD 0 (USD 0 over the target). The target is a hypothetical control target, not a limit. Main cost drivers and savings worth trying:

- No line costs money: every module is written in-house under MIT and every bought part is free and open source. The two added lines (the repository reader and the 3D viewer script) are also USD 0.
- The real costs to a user are install size and CI time. Splitting the dependencies (RDK-DDR-003, P3) cuts the install for a check from about 1.1 GB to about 5 MB, and the per-repository stub (P2) cuts the kit carried by each repository from 1.20 MB to about 47 kB.
- Savings worth trying: cache the extras between CI runs on release tags; publish the template so setup needs no clone of a finished repository.
- The only paid option is private-repository CI time beyond GitHub's included minutes, which the tool does not need.

■ **Decisions made**

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	D1 distribution as a pip package plus a template repository; D2 identity in <code>readykit.yaml</code> ; D3 add ISO A3 sheets; D4 one <code>readykit</code> command; D6 Open Know-How export; D7 TRL 3 evidence for a software project; D8 add at least six missing checks (R6); D9 write the getting-started guide with the package (R15)	Amish: "i accept all your recommendations, go with them across all repos."	RDK-DDR-001, RDK-DDR-002
2026-09-25	Hold TRL 4 work across the portfolio	Amish: "you know the drill, nothing gets past TRL 3"	RDK-DDR-001
2026-09-26	D5 reversed: keep the portfolio license pair, CERN-OHL-S v2 for hardware and content, MIT for software	Amish: "apply the same MIT license pairing across the board for all repos."	RDK-DDR-002
2026-09-30	Make every design constructable while drawing the build plan, and keep outstanding decisions out of the plan, in this register	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." and "don't log outstanding decisions in this build plan - that is not the place for it."	RDK-DDR-003 (changes P1 to P6 made under this instruction; accepted on 2026-10-02, below)
2026-10-01	Budgets are value-engineering targets, not limits	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register, Value engineering
2026-10-02	Name: keep "ReadyKit"; recheck the package index just before the first release and publish the first real release under that name as soon as the package installs (the GitHub repository already exists under Amish's account)	Amish: "i approve your recommendations for all 555 open decisions."	RDK-DDR-001, 01
2026-10-02	Pilot users: an open science hardware group, with GOSH (Gathering for Open Science Hardware) as the first candidate, recruiting two or three member projects through its community forum; a university lab as the second pilot	Amish: "i approve your recommendations for all 555 open decisions."	RDK-DDR-001, 02
2026-10-02	Windows: supported through WSL2 for the first release; native Windows added later if pilot users ask for it	Amish: "i approve your recommendations for all 555 open decisions."	RDK-DDR-001, 03
2026-10-02	Decision-wording rule (R13): generic, so any decision written as made must name an owner and a date; each repository sets its 'still open' phrase in its identity file, defaulting to "Proposed, awaiting"	Amish: "i approve your recommendations for all 555 open decisions."	RDK-DDR-001, 04
2026-10-02	Scope of the first package (option a): release gate and archive helper in the release extra,		RDK-DDR-003, A1

Date	Decision	Decided by	Record
	build plan pictures in the media extra; photoreal renders and storefront cards kept for the portfolio only	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-02	Oldest Python supported: 3.11 or later (option b), tested in CI on 3.11 and the newest release; 3.10 reaches end of life in October 2026. This differs from the register's earlier recommendation (a)	Amish: "i approve your recommendations for all 555 open decisions."	RDK-DDR-003, A2
2026-10-02	Design for construction accepted: the changes P1 to P6, as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	RDK-DDR-003 , Table 1